Tang Song Soundscape Project Report

Online Visualization Link: [Tang Song Soundscape](). Source Code Link [Here]().

1. Introduction and Domain Problem

Project Background: Currently, the public primarily experiences classical Chinese literature by reading the original texts. Most analyses focus on visual "imagery," such as isolated scenic elements. However, sound descriptions are also crucial for building the atmosphere of a poem. Much like how modern films use sound to set a mood and reflect a character's state of mind, auditory details in classical poetry are essential for creating a deep sense of immersion.

Domain Problem: This project aims to transform abstract sound descriptions in Tang and Song poetry—such as cicadas, flutes, and rain—into intuitive visual maps. By mapping these sounds, we can explore how the locations and types of auditory imagery shifted over 600 years.

Goal and Significance: The goal is to explore historical "soundscapes" to help readers better empathize with poets across time and space. By mapping these sounds geographically, this project provides a panoramic, cross-temporal perspective on the literary works of the Tang and Song dynasties. This approach allows us to see how the political climate and regional changes influenced the sounds that poets recorded in their work.

2. Dataset and Data Abstraction

Data Acquisition and Preprocessing: The primary datasets for this project are *Three Hundred Tang Poems* and *Three Hundred Song Ci*, sourced from an existing JSON database on GitHub*(chinese-poetry, n.d.)*. After converting the texts from Traditional to Simplified Chinese, I built a foundational "poetry ontology database." This database stores basic metadata for each text, including Poem ID, Dynasty, Genre, Author, and Title.

Sound Event Database and Validation: Next, I constructed a "Sound Entry Level" database to analyze the specific auditory details. I used a LLM to extract the data based on several key dimensions:

- Event Details: The original phrase, the source of the sound, the context window, and its category.
- Perception Modality: How the sound was perceived. This is divided into three types: explicitly "heard" (using a verb), inferred (e.g., bird songs without a listening verb), and implied (e.g., a strong wind blowing).
- Perception Reality: Whether the sound was real, or if it occurred in a dream, memory, or imagination.

- Absence of Sound: Marking negated auditory states, such as “soundless” or “unheard.”

Data Validation: To prevent LLM hallucinations, I first created a Minimum Viable Product (MVP) using 20 manually annotated poems to test the LLM’s accuracy. After processing the full dataset, I manually filtered the results based on the LLM’s confidence scores to ensure high accuracy, aligning with the principles of Human-Centered AI to ensure reliable and trustworthy automation (Shneiderman, 2022).

Geocoding Pipeline: Matching historical sound events to specific geographic coordinates was the most complex step. I designed a 4-level priority pipeline to maximize accuracy:

- Priority 1: Direct Mentions. Locations explicitly written in the poem body (e.g., specific cities or landmarks). These have the highest accuracy.
- Priority 2: Title or Preface. Locations mentioned in the title or the introduction of the poem.
- Priority 3: Web Scraping. I wrote a web crawler to cross-reference unmatched poems with the Souyun website **(**Wang, n.d.**)**’s chronological poetry map.
- Priority 4: Human & LLM Verification. For remaining locations, I used two different LLMs (DeepSeek, n.d. and Google, n.d.) and manual research (using historical commentaries and the poets’ biographies) to reach a consensus.

Out of more than 700 extracted sound events, I successfully determined 644 geographic locations. About 30 of these were estimated by later historians; to ensure data integrity, I explicitly marked their certainty level as “F” (False/Uncertain) in the database.

3. Task Abstraction

Guided by the nested model for visualization design and validation (Munzner, 2009) and based on a multi-level typology of visualization tasks (Brehmer & Munzner, 2013), I broke down the high-level domain problem into executable low-level tasks for users:

- **T1.** Observe historical spatio-temporal shifts: Use the dynasty-switching feature to observe how the geographical distribution of sound descriptions shifted across different historical periods.
- **T2.** Observe the distribution of sound categories: Use the filtering function to explore how different types of sounds are distributed across various dynasties and regions.
- **T3.** Explore textual details in specific regions: Click on a specific city or area to deeply explore the original poetic texts and historical contexts of the sounds mentioned there.
- **T4.** Identify high-density clusters: Use the KDE Heatmap to quickly identify and

locate the "noisiest" areas on the map, where auditory imagery is most concentrated.

- **T5.** Compare poets' preferences: Select specific poets to compare their sound description preferences using a unified data metric.

4. Design Rationale: Visual Representation and Interaction

This section explains the design choices for the visualization idioms and interactions, mapped to the tasks (T1-T5) defined in the previous section.

Map Tab (*Supports Tasks: T1, T2, T3, T4)*

Visual Encoding:

- *Position (T1, T2, T4):* I used 2D spatial position (latitude and longitude) as the primary visual channel to encode the geographical coordinates of sound events. According to visual perception rankings, position on a common scale is the most accurate channel for human perception of geospatial distribution. According to visual perception studies, position on a common scale is the most accurate channel for human perception of data distribution (Heer & Bostock, 2010).
- *Dynasty Distinction (T1):* To represent the shift between historical periods, I used distinct territory shapes (Polygons) and background colors, combined with a dynasty-switching toggle.
- *Color Hue (T2):* Sound categories (Animal, Nature, Human, Instrument, Environment) are encoded using five distinct colors derived from traditional Chinese painting (e.g., Gamboge, Stone Green). Color hue is an effective channel for categorical data.
- *KDE Heatmap (T4):* To identify "noisy" clusters, I implemented Kernel Density Estimation (KDE) to generate an "ink-wash" style heatmap. This allows users to intuitively perceive the shifting "geographic center of gravity" of sounds across different eras.

Interaction Design:

- *Marker Clustering (T3):* To solve the problem of data congestion in dense areas, I used Marker Cluster technology that automatically groups or separates points based on the zoom level.
- *Details-on-Demand (T3):* To avoid visual clutter and overlapping text, I adopted a "details-on-demand" strategy (Yi et al., 2007; Dimara & Perin, 2020). When a user clicks a point, a dedicated information panel displays the original poem, bilingual context, and background metadata.

Poets Tab (*Supports Tasks: T5, T3)*

Visual Encoding:

- *Radar Chart (T5):* I designed a proportional weight radar chart to compare poets' preferences. Based on the "Expressiveness Principle," I used percentages rather than absolute counts. This ensures a "fair competition" in visual comparison between highly prolific poets (e.g., Su Shi) and those with fewer surviving works.

Interaction Design:

- *Cross-Page Linkage (T3):* To achieve a deep binding of "Poetry, Place, and Person," I implemented a cross-page interaction. Clicking on a specific sound phrase in the poet's list or the dynamic word cloud automatically navigates the user back to the map and zooms into the exact geographical coordinate where that sound occurred.

5. Iterative Process

This project underwent several rounds of iteration across data processing, technology selection, and user experience design.

Data Construction Iteration: Initially, I intended to extract and map "emotions" from the poems. However, classical Chinese poetry is highly nuanced (e.g., using joyful scenery to express grief), making direct emotion extraction via LLMs unreliable. Consequently, I pivoted to extracting "sounds," which are more objective. To mitigate the risk of LLM hallucinations during extraction, I introduced a "human-in-the-loop" mechanism, manually checking and refining the dataset to ensure high accuracy.

Geocoding Pipeline Iteration: Matching poems to specific geographical locations proved highly challenging due to vague historical references. To solve this, I established a strict, prioritized verification pipeline.

Technology and Platform Iteration: I initially planned to build the final visualization entirely within QGIS. However, I realized that QGIS could not support the complex interactive features I envisioned, such as details-on-demand and cross-page linkage. Therefore, I adapted my methodology: I used QGIS to process the spatial data and generate GeoJSON files, and then shifted to deploying a custom web frontend. To achieve this efficiently, I utilized an LLM (Anthropic, 2024) to write the majority of the web application's codebase, while I manually adjusted the specific design details, interactions, and visual logic.

User Study and Design Improvements:

Based on feedback from qualitative user testing, I implemented several crucial design iterations:

- Visual Encoding Refinement: User testing revealed that displaying absolute counts on the radar chart was visually misleading when comparing poets with vastly different output volumes. I changed the data metric from absolute counts to percentages. This strictly adheres to the Expressiveness Principle in data visualization. Additionally, I adjusted the legend and category colors to better match the map's overall aesthetic and improve readability.
- Cognitive Load Optimization: Participants were confused by the English abbreviations "N. Song" and "S. Song" (mistaking them for Pinyin initials). I replaced these with clear Chinese labels ("北宋/南宋") to eliminate cross-lingual cognitive ambiguity.
- Performance Optimization: Users experienced severe rendering lag when zooming into highly dense data clusters on the map. I significantly resolved this performance issue by disabling complex SVG filters during zoom actions and enabling Leaflet's preferCanvas rendering mode.

6. Lessons Learned

User testing revealed a stark disconnect between the designer's and the user's mental models. Design choices I considered "obvious"—such as English abbreviations and cluster numbers—caused unexpected confusion. This practical experience perfectly validates Norman's (2013) philosophy regarding mental model mismatches.

Matching historical humanities data to modern geography is highly complex. Open-source historical shapefiles are often fragmented because they only map areas with "absolute historical certainty". This limitation forced me to manually vectorize historical map images into polygon layers to accurately represent actual spheres of influence.

I initially considered adding extra features, such as hover tooltips for exact record counts. However, prototyping showed these additions felt redundant and cluttered the UI. This reinforced that avoiding "feature creep" is crucial to keeping the visualization efficient and focused.

Lacking an ideal historical basemap, I found that using pure topographical maps provided geographical context but ruined the traditional artistic aesthetic. I learned that visualization requires calculated compromises; I prioritized visual clarity and artistic immersion over excessive background detail.

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